Walchand College of Engineering, Sangli
Computer Science & Engineering
Third Year

Course: Computer Algorithm Lab

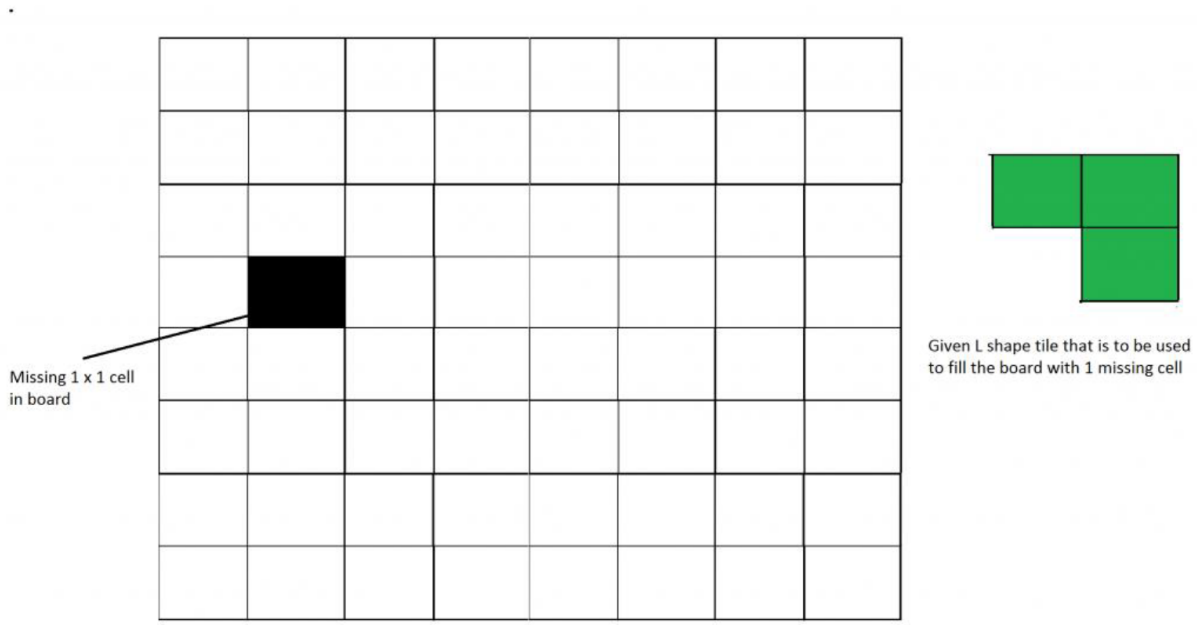
Week 4 Assignment

Part: 1

Divide and conquer strategy

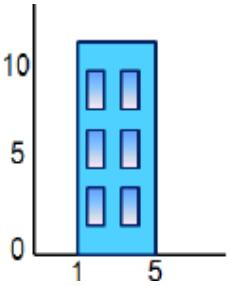
Q1) Implement algorithm to Find the maximum element in an array which is first increasing and then decreasing, with Time Complexity O(Logn).

Q2) Implement algorithm for Tiling problem: Given an n×n board where n is of form 2^k where k >= 1 (Basically n is a power of 2 with minimum value as 2). The board has one missing cell (of size 1 x 1). Fill the board using L shaped tiles. An L shaped tile is a 2 x 2 square with one cell of size 1×1 missing

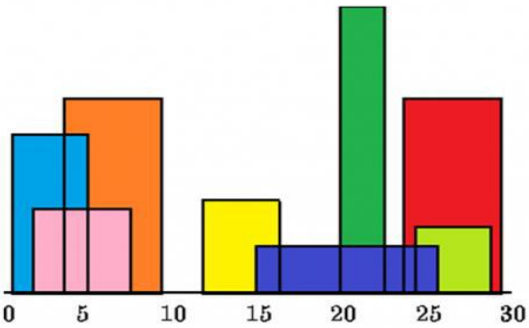


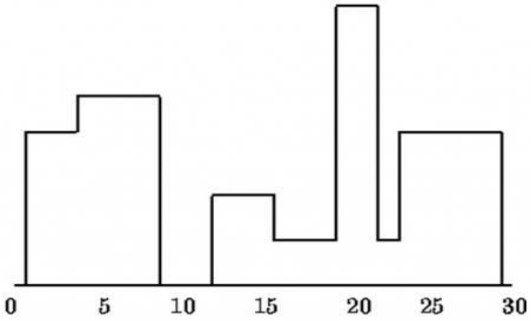
Q3) Implement algorithm for The Skyline Problem: Given n rectangular buildings in a 2-dimensional city, computes the skyline of these buildings, eliminating hidden lines. The main task is to view buildings from a side and remove all sections that are not visible.

All buildings share common bottom and every building is represented by triplet (left, ht, right) ‘left’: is x coordinated of left side (or wall). ‘right': is x coordinate of right side ‘ht': is height of building. For example, the building on right side is represented as (1, 11, 5)

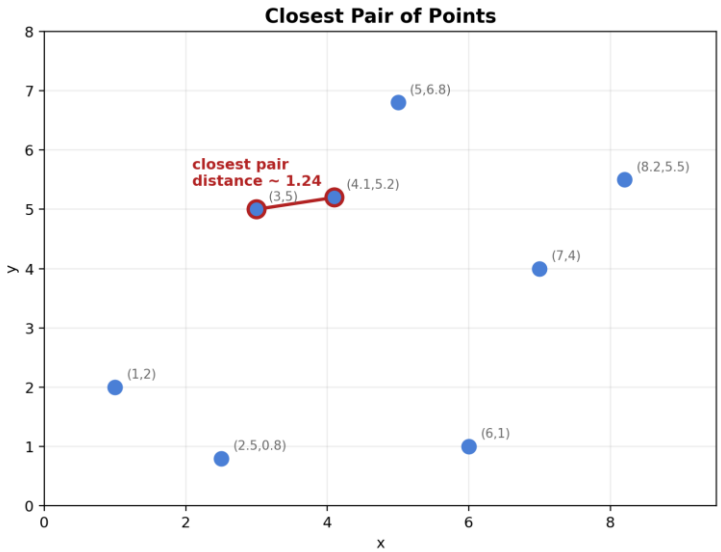


A **skyline** is a collection of rectangular strips. A rectangular **strip** is represented as a pair (left, ht) where left is x coordinate of left side of strip and ht is height of strip.
With Time Complexity O(nLogn)





Q4) Implement an algorithm using Divide and Conquer to find the pair of points with the smallest Euclidean distance among n points given in a 2-dimensional plane.
Given the (x, y) coordinates of n points, find the two points that are closest to each other and report the minimum distance between them. With Time Complexity O(nLogn)



Q5) Implement Strassen's algorithm using Divide and Conquer to multiply two n x n matrices A and B, where n is a power of 2, using fewer recursive sub-multiplications than the straightforward block-multiplication approach. With Time Complexity O(n^log2(7)), i.e. approximately O(n^2.81)

Note: You may refer to similar problems on platforms such as LeetCode or CodeChef to understand the problem statement, but your implementation, choice of pivot/split strategy, and analysis must be your own. Treat the sample inputs above only as illustrations of the expected input and output format — your program should work correctly for any valid input of these types, not just the example shown. At least 4 test case should be available in the output document.